

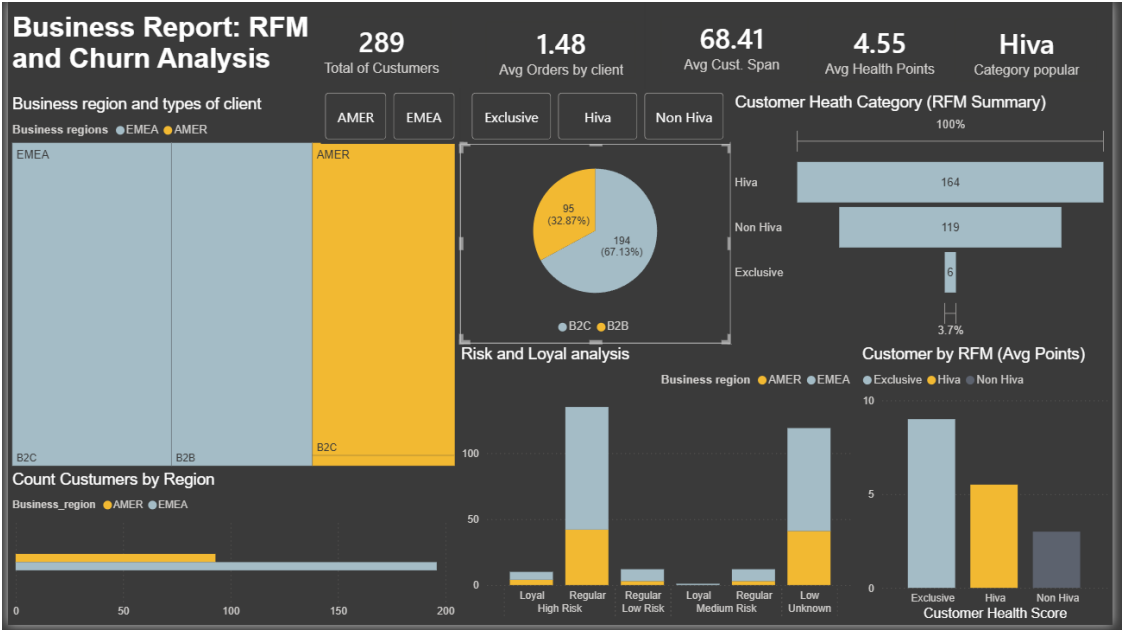
Customer Retention Hub: SQL - RFM Segmentation, Churn & Loyalty Analysis

End-to-end data engineering & BI project built with MySQL and Power BI

Project repository: [SQL-MySQL Project \(Onedrive\)](#)

Project Summary

Designed and built an end-to-end relational data ecosystem using **MySQL (HeidiSQL)** and **Power BI**, automating advanced customer segmentation through RFM profiling, loyalty tiering, and churn-risk analytics. The system turns raw transactional logs into actionable retention strategies for marketing and operations teams.



Executive Power BI dashboard — RFM & Churn Analysis (289 customers, segmented by region and loyalty tier)

Key Results

289	1.48	68.4	180+ days
Total customers analyzed	Average orders per customer	Average customer lifespan (days)	Automated churn-risk alert threshold

1. Business Need

Transactional data was fragmented across isolated tables, making it impossible to identify high-value accounts or detect early signs of customer attrition. Marketing and operations teams lacked a centralized view to launch targeted retention campaigns. The business needed an automated, scalable data framework to grade customer health, forecast churn risk, and calculate lifecycle spans without relying on manual queries.

2. Strategic Importance & Decision-Making Impact

This project transforms historical data into predictive intelligence with three core impacts:

- ▶ **Proactive churn mitigation:** dynamically identifies "high-risk" accounts (180+ days inactive) to trigger automated win-back campaigns before customer loss becomes permanent.
- ▶ **Health grading & segmentation:** applies a custom RFM matrix to tier customers (Non Hiva, Hiva, Exclusive), enabling precise budget allocation for high-value accounts and loyalty perks.
- ▶ **Operational agility:** tracks customer activity spans and purchase frequency, helping sales and supply chain teams accurately forecast demand and optimize communication.

3. Technology Stack & BI Automation

To ensure data integrity, query performance, and automated reporting, I deployed an integrated SQL and BI stack:

- ▶ **Database engineering (MySQL & HeidiSQL):** built a robust relational schema with strict foreign keys and CHECK constraints mapping Customers, Orders, Items, and Bills.
- ▶ **Optimized pipelines:** developed modular SQL scripts and performance-optimized database views (v_customer_features) to standardize metric formulas and isolate workloads.
- ▶ **Advanced RFM logic:** leveraged CTEs (WITH clauses) and programmatic CASE constructs to calculate multi-dimensional Recency, Frequency, and Monetary scores.
- ▶ **Business Intelligence (Power BI):** built interactive executive dashboards with drill-down by region (EMEA, AMER), client type (B2B, B2C), and risk segment.

Technical Skills Demonstrated

Advanced SQL	CTEs (WITH), window functions, programmatic CASE logic, optimized views, foreign keys & CHECK constraints
Data modeling	Relational schema design (ERD), normalization, referential integrity
Business Intelligence	Power BI: interactive dashboards, drill-down, dynamic segmentation
Business analytics	RFM segmentation, churn scoring, retention & loyalty KPIs

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